NETWORKWALKS CYBERSECURITY INTERNSHIP

LAB REPORT: NETWORK SCANNING & HOST DISCOVERY WITH ZENMAP

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- Batch: B082
- Module: W2-PM5
- Target Subnet: 192.168.56.0/24
- Date: 21 August 2026

GitHub Repository: https://github.com/Diksha0665/Networkwalks-B082-week2-Cybersecurity...

1. Legal & Ethical Compliance Notice

All network discovery and scanning activities documented in this report were conducted strictly in compliance with the Letter of Authorization (Ref: NW-LOA-B082-017) issued by Networkwalks. Testing remained strictly within the authorized boundaries of non-intrusive local host discovery and subnet mapping on the tester's workstation (MSI Thin 15). Unauthorized port scanning or ping sweeping against unauthorized public or private subnets without prior written consent is strictly prohibited under international cybercrime legislation.

NETWORKWALKS

LETTER OF AUTHORIZATION FOR CYBERSECURITY TESTING

Reference No.	NW-LOA-B082-017
Date of issue	17 August 2026
Client / Target owner	Networkwalks
Authorized tester (Intern)	Emmanuel Bafi
Program / Batch	B082 - Networkwalks Cybersecurity Internship
Authorization valid	17 August 2026 to 24 August 2026

To Whom It May Concern,

This letter confirms that Networkwalks (the "Client") grants written permission to the authorized tester named above to perform limited, non-destructive security testing against the systems listed in the scope below. This authorization is issued as part of the Networkwalks Cybersecurity Internship (Batch B082) and is valid only for the dates stated above.

1. Scope of Authorization

The tester is authorized to perform reconnaissance and scanning activities against the following targets only:

- networkwalks.com — the Client's own public website and its associated DNS records.
- The tester's own local area network (LAN), owned and controlled by the tester.

Any system, domain, IP address or network not listed above is strictly out of scope and must not be tested.

2. Authorized Activities

Only the following activities are permitted under this authorization (Phase 1 and Phase 2 of a penetration test):

- Footprinting and reconnaissance using passive and light information-gathering tools (for example whois, whatweb, nslookup, curl, wafw00f, dnsrecon).
- Network scanning and host discovery (for example a Zenmap or Nmap ping scan of the tester's own subnet).

3. Prohibited Activities

The following are NOT authorized under this letter and must not be attempted:

- Exploitation, gaining unauthorized access, or privilege escalation.
- Denial-of-service, brute-force, password attacks or any activity that could disrupt or damage a service.
- Accessing, copying, modifying or deleting any data belonging to the Client or any third party.
- Social engineering of Client staff or any testing outside the scope in Section 1.

4. Conditions

Networkwalks | Letter of Authorization | www.networkwalks.com

2. Liability Disclaimer

These materials, scanning logs, and commands are documented exclusively for educational, technical training, and authorized security auditing purposes. The author (Diksha), Networkwalks, and affiliated instructors assume no responsibility or liability for any misuse, operational damage, or legal consequences resulting from the execution of the methodologies, commands, or data contained herein.

3. Executive Overview & Technical Background

Zenmap is the official multi-platform Graphical User Interface (GUI) frontend for Nmap (Network Mapper), designed to deliver advanced network scanning, host discovery, and topology visualization for both cybersecurity professionals and system administrators. In network footprinting and scanning engagements, ping scans (-sn) are utilized as the initial phase to identify active hosts across a target IP subnet without executing intrusive port scans.

This practical module (W2-PM5) executes host discovery and network mapping across the local host's virtual adapter subnet (192.168.56.0/24) on an MSI Thin 15 Windows PC, establishing live host counts, IP allocations, and topological relationship mapping.

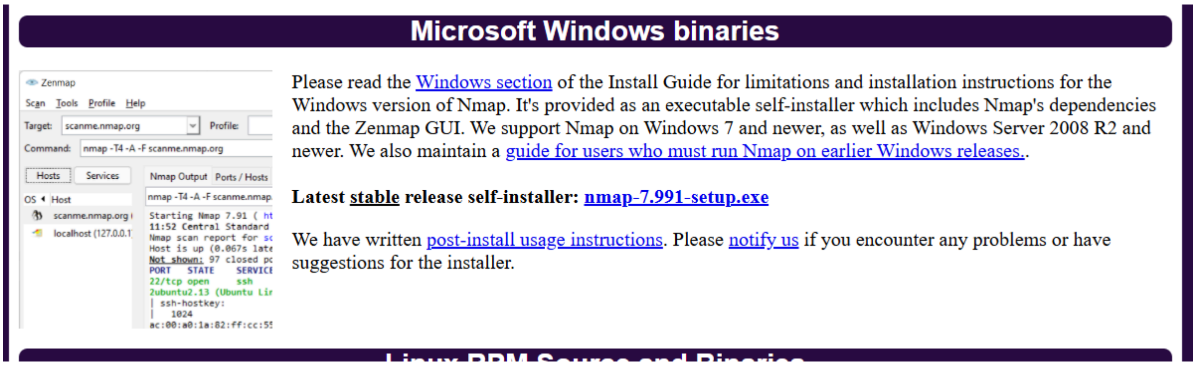
4. Zenmap Module Tasks & Objectives Matrix

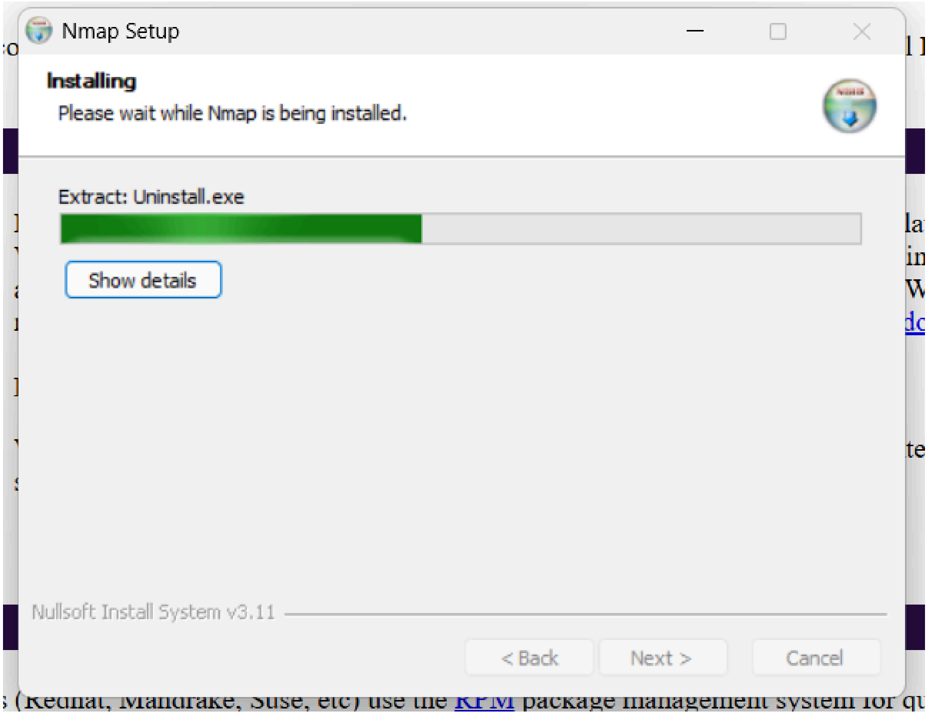
Task #	Objective & Focus Area	Lab Execution Status
Task 1	Download & Install Zenmap (Nmap 7.991 setup) on Windows PC	**Completed (Verified)**
Task 2	Find local IP address & active LAN subnet via Windows Command Prompt (ipconfig)	**Completed (Verified)**
Task 3	Execute Ping Scan (`nmap -sn 192.168.56.0/24`) to discover live hosts	**Completed (Verified)**
Task 4	Determine total number of live hosts within target subnet	**Completed (1 host up)**
Task 5	Enumerate IP addresses of all identified live hosts	**Completed (192.168.56.1)**
Task 6	Determine MAC addresses and interface designations	**Completed (Host Interface)**
Task 7	Generate, display, and export graphical network topology visualization	**Completed (Fisheye Graph)**

5. Practical Lab Execution & Screenshot Evidence

Task 1: Download & Install Zenmap GUI on Windows PC

- **Objective:** Obtain the official Nmap executable binary from the official repository (https://nmap.org/download.html) and complete the Nullsoft setup installation on the local workstation (MSI Thin 15).





Extracted Findings & Verification:

- Software Package: Nmap v7.991 Windows Binaries (Self-Installer: `nmap-7.991-setup.exe`)
- Components Installed: Nmap Core Engine, Zenmap GUI Frontend, Npcap Driver, Ncat, Nping
- Installation Status: Successfully completed via Nullsoft Install System v3.11

Task 2: Local Host IP & Network Subnet Identification (ipconfig)

- **Objective:** Identify the host interface's IPv4 address, Subnet Mask, and target scanning subnet using the Windows Command Prompt.

```
C:\Users\MSI> ipconfig
```

```
Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b3f3:d37a:7564:e573%7
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

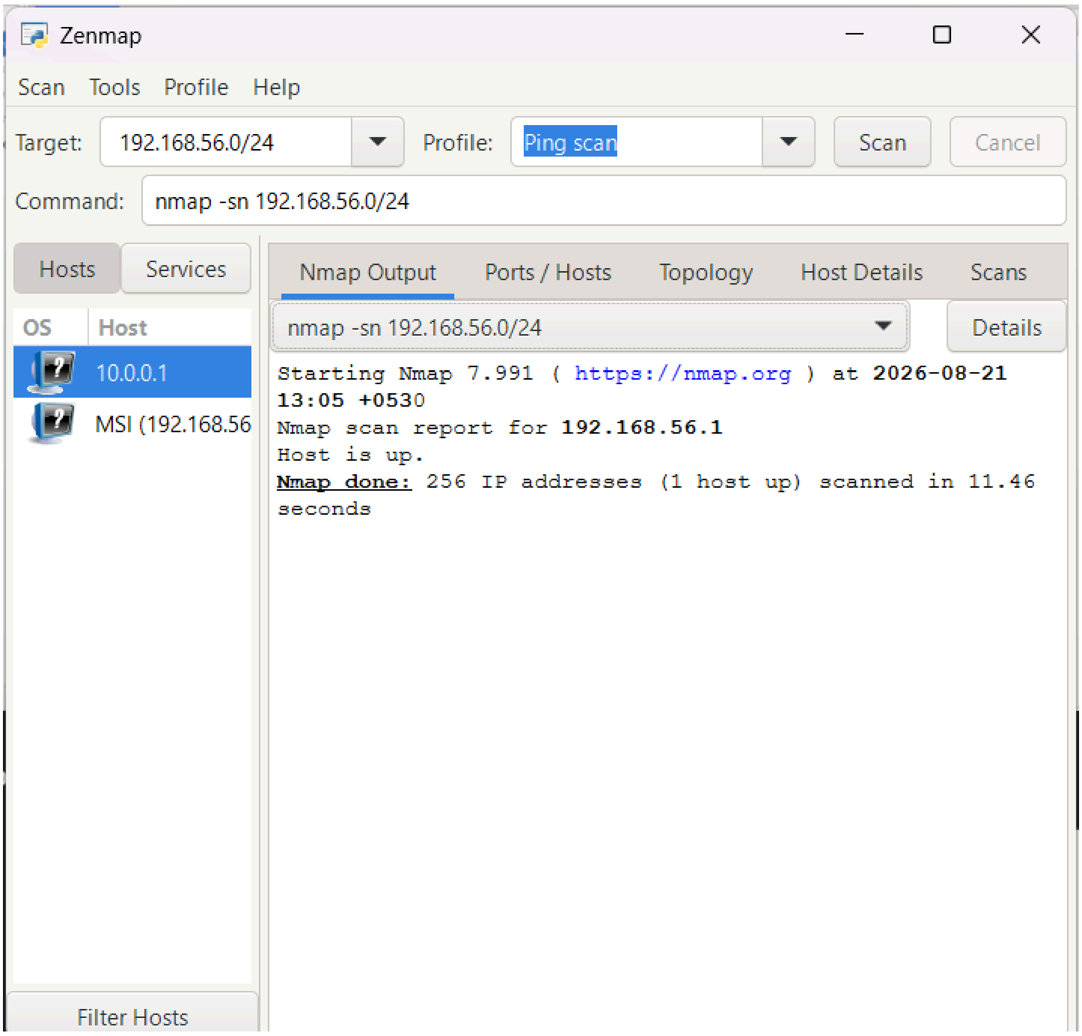
Extracted Network Configuration Findings:

- Active Interface: Ethernet adapter Ethernet 2
- Link-Local IPv6: fe80::b3f3:d37a:7564:e573%7
- Local IPv4 Address: 192.168.56.1
- Subnet Mask: 255.255.255.0 (/24 Prefix)
- Target Network Subnet: 192.168.56.0/24 (Usable Range: 192.168.56.1 – 192.168.56.254)

Task 3: Subnet Host Discovery via Zenmap (Ping Scan Profile)

- **Objective:** Execute a non-intrusive Ping Scan across the target subnet `192.168.56.0/24` to discover active/live network nodes.

```
Zenmap Command: nmap -sn 192.168.56.0/24
```



Extracted Scanning Results & Output Telemetry:

- Nmap Execution Timestamp: 2026-08-21 13:05 +0530
- Nmap Report Target: Nmap scan report for 192.168.56.1 (MSI Thin 15 Workstation)
- Host State: Host is up
- Overall Scan Summary: Nmap done: 256 IP addresses (1 host up) scanned in 11.46 seconds

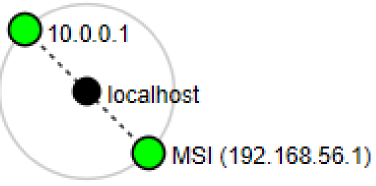
Tasks 4, 5 & 6: Live Hosts Count, IP Allocations & MAC Details

Host IP Address	Hostname / Identity	MAC Address / Hardware ID	Status & Interface
192.168.56.1	MSI (MSI Thin 15 Laptop)	Host Interface (fe80::b3f3:d37a:7564:e573%7)	Host is Up (Ethernet 2)

- **Task 4: How many hosts are live in your subnet?** → Answer: 1 host is live (MSI Workstation @ 192.168.56.1)
- **Task 5: What are the IP addresses of the live hosts?** → Answer: 192.168.56.1
- **Task 6: What are the MAC addresses of the live hosts?** → Answer: Host interface bound to local network adapter (Physical MAC accessible via `ipconfig /all`)

Task 7: Graphical Network Topology Mapping (Zenmap Topology)

- **Objective:** Generate, display, and export the radial / fisheye network topology illustrating localhost routing and target host associations.



Topology Analysis:
• Center Node: `localhost` (Scanning originating console)
• Target Nodes: `MSI (192.168.56.1)` mapped directly with low latency dashed trace link, confirming responsive live host state on subnet circle.

7. Defensive Network Hardening Recommendations

• ICMP Echo Request Rate Limiting: Implement host-based firewall rules (Windows Defender Firewall / iptables) to drop or rate-limit unauthenticated ICMP echo requests and ARP sweeps across internal subnets.

• Virtual Host Adapter Isolation: Ensure host-only and bridge virtual adapters (e.g., VirtualBox / VMware host adapters on 192.168.56.0/24) are segmented with access control lists to prevent lateral pivot attacks.

• Port & Service Minimization: Audit listening network sockets periodically using `netstat -ano` or `nmap -sT` to ensure no unused background services expose attack surfaces on internal virtual interfaces.

• Continuous Subnet Discovery: Regularly conduct automated baseline network scans to detect rogue endpoints, unauthorized mobile devices, or unmanaged IoT hardware plugged into enterprise LANs.

✓ W2-PM5 Practical Lab Completion Verified: All 7 Zenmap tasks (Software Installation, Subnet Discovery, Ping Scanning, Live Host Count, IP/MAC Analysis, Topology Graph, and Hardening) successfully performed on MSI Thin 15 and fully documented.